

SIMONE ROMITI

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Computational physicist with a PhD and 4+ years of postdoctoral research experience in Monte Carlo simulations, Bayesian inference, and high-performance computing. Production-grade C++ and Python developer; main author and maintainer of open-source scientific libraries with CI/CD and versioned releases. Principal Investigator on two multi-million GPU-hour allocations (CSCS ALPS 2025, GCS JUPITER 2026). Comfortable owning numerical methods end-to-end: derivation, implementation, statistical validation, and uncertainty quantification on terabyte-scale simulation outputs.

TECHNICAL SKILLS

Languages - C/C++ (production, templates, modern C++), Python (production, packaging), R, Bash/GNU Linux

Numerical & Statistical Methods - Monte Carlo, Markov Chain Monte Carlo, Hybrid/Hamiltonian Monte Carlo, Bayesian inference, Nested Sampling, uncertainty quantification, FFT-based spectral methods, constrained optimization

Machine Learning - PyTorch, Physics-Informed Neural Networks (PINNs), score-based Diffusion Models

Scientific Python - NumPy, SciPy, Pandas, Matplotlib, Plotly, Streamlit, SymPy, Jupyter

HPC & Systems - CUDA, MPI, OpenMP, SLURM, EasyBuild

DevOps & Tools - Git, GitHub Actions, CI/CD, Docker, LaTeX, Quarto

Spoken Languages - Italian (native), English (proficient), German (basic)

WORK EXPERIENCE

University of Bern

Postdoctoral Researcher

Apr 2024 – Present

Bern, Switzerland

- **Reduced algorithmic complexity from $O(N^6)$ to $O(N \log N)$** for a class of high-dimensional integrals by reformulating the kernel via parallelized Fast Fourier Transforms.
- Built a **Bayesian inference pipeline** processing observables computed on $\sim 10^3$ simulation configurations (~ 1 TB each); features uncertainty quantification, systematic error budget, and model averaging. Released as open-source library.
- Developed a **Physics-Informed Neural Network** approach to the Schrödinger equation with periodic potentials (sole-author publication, open-source implementation).
- Applied **Nested Sampling** to multi-modal density estimation for the study of phase transitions.
- **Principal Investigator** on two large compute allocations totalling $\sim O(10^6)$ GPU-hours (CSCS ALPS 2025, GCS JUPITER 2026).
- Mentoring of multiple Master's and PhD students.

University of Bonn

Postdoctoral Researcher

Nov 2021 – Mar 2024

Bonn, Germany

- Designed and operated **large-scale Monte Carlo simulations** on international HPC clusters: parameter tuning, statistical validation, autocorrelation analysis, and quality control over ensembles.
- Derived and implemented a **numerical method achieving machine-precision accuracy** for a class of constrained optimization problems.
- Co-developed a **hybrid Monte Carlo + quantum-computing** method for estimating high-dimensional integrals.

SELECTED OPEN-SOURCE PROJECTS

- **[lattice_data_tools](#)** – Python toolkit for the production and statistical analysis of lattice simulation data. Main author and maintainer.
- **[su2](#)** – C++ templated library for Monte Carlo simulation of SU(N) theories with OpenMP parallelization. Main editor.
- **[tmLQCD](#)** – Large-scale C library for lattice simulations on GPU clusters (CUDA + MPI). Contributor.
- **[adiabatic_PINNs-suN](#)** – Physics-Informed Neural Network solver for the Schrödinger equation in SU(N) lattice gauge theory; reference implementation for sole-author publication.

EDUCATION

PhD in Physics | title awarded in April 2022 |

Roma Tre University | Nov. 2018 – Oct. 2021

M.S. in Physics | Grade: 110/110 *cum laude* | GPA: 29.85/30 |
B.S. in Physics | Grade: 110/110 *cum laude* | GPA: 28.84/30 | Scholarship

Roma Tre University | Oct. 2016 – Oct. 2018
Roma Tre University | Oct. 2013 – Jul. 2016

LEADERSHIP & COMMUNICATION

Invited Speaker – Technical seminars at [CERN](#) and [ECT*](#)
Workshop Organizer – International workshop on HPC & Quantum Computing

2025 – 2026
[ECT*](#) | 2025